

46 YEARS OF EXCELLENCE | NARAYANA IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX) **GTM-19** **Date: 17.01.2025**
Time: 3.00Hrs. **JEE MAIN MODEL** **Max. Marks: 300**

PHYSICS MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A train is moving with a speed of 12 m/s on rails which are 1.5 m apart. To negotiate a curve of radius 400 m, the height by which the outer rail should be raised with respect to the inner rail is

(Given, $g = 10\ \text{m/s}^2$)

A) 5.4 cm B) 4.8 cm C) 6.0 cm D) 4.2 cm

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2. An electric charge $10^{-6}\mu C$ is placed at origin $(0,0)m$ of $X-Y$ co-ordinate system. Two points P and Q are situated at $(\sqrt{3},\sqrt{3})m$ and $(\sqrt{6},0)m$ respectively. The potential difference between the points P and Q will be

A) $\sqrt{3}V$ B) $\sqrt{6}V$ C) 3 V D) 0 V

3. If the refractive index of the material of a prism is $\cot\left(\frac{A}{2}\right)$, where A is the angle of prism then the angle of minimum deviation will be

A) $\pi-2A$ B) $\frac{\pi}{2}-A$ C) $\frac{\pi}{2}-2A$ D) $\pi-A$

4. A rectangular loop of length $2.5\ m$ and width $2\ m$ is placed at $60°$ to a magnetic field of $4T$. The loop is removed from the field in $10\ s$. The average emf induced in the loop during this time is

A) +2V B) +1V C) –2V D) –1V

5. Given below are two statements :

Statement (I) : Viscosity of gases is greater than that of liquids.

Statement (II) : Surface tension of a liquid decreases due to the presence of insoluble impurities.

In the light of the above statements, choose the most appropriate answer from the options given below :

A) Statement I is correct but Statement II is incorrect

B) Statement I is incorrect but Statement II is correct

C) Both Statement I and Statement II are correct

D) Both Statement I and Statement II are incorrect

6. $0.08\ kg$ air is heated at constant volume through $5°C$. The specific heat of air at constant volume is $0.17\ kcal/kg°C$ and $J=4.18 joule/cal$. The change in its internal energy is approximately.

A) 142 J B) 298 J C) 284 J D) 318 J

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7. Give below are two statements :

Statement (I) : Planck's constant and angular momentum have same dimensions.

Statement (II) : Linear momentum and moment of force have same dimensions.

In the light of the above statements, choose the correct answer from the options given below :

A) Statement I is true but Statement II is false

B) Both Statement I and Statement II are false

C) Statement I is false but Statement II is true

D) Both Statement I and Statement II are true

8. Identify the physical quantity that cannot be measured using spherometer :

A) Thickness of thin plates

B) Specific rotation of liquids

C) Radius of curvature of convex surface

D) Radius of curvature of concave surface

9 The acceleration due to gravity on the surface of earth is g. If the diameter of earth reduces to half of its original value and mass remains constant, then acceleration due to gravity on the surface of earth would be :

A) $\frac{g}{2}$ B) $\frac{g}{4}$ C) $2g$ D) $4g$

10. A body of mass $1000\ kg$ is moving horizontally with a velocity $6\ m/s$. If $200\ kg$ extra mass is added, the final velocity (in m/s) is :

A) 6 B) 5 C) 2 D) 3

11. A convex lens of focal length $40\ cm$ forms an image of an extended source of light on a photoelectric cell. A current I is produced. The lens is replaced by another convex lens having the same diameter but focal length 20 cm. The photoelectric current now is :

A) $4I$ B) $2I$ C) I D) $\frac{I}{2}$

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12 A plane electromagnetic wave propagating in x-direction is described by $E_y = (200\ Vm^{-1})\sin[1.5\times10^7 t - 0.05x]$; The intensity of the wave is :

(Use $\in_0 = 8.85\times10^{-12} C^2N^{-1}m^{-2}$)

A) 26.6 Wm^{-2} B) 53.1 Wm^{-2} C) 106.2 Wm^{-2} D) 35.4 Wm^{-2}

13. A wire of resistance R and length L is cut into 5 equal parts. If these parts are joined parallelly, then resultant resistance will be

A) 5 R B) 25 R C) $\frac{1}{5}R$ D) $\frac{1}{25}R$

14. A proton moving with a constant velocity passes through a region of space without any change in its velocity. If $\vec{E}$ and $\vec{B}$ represent the electric and magnetic fields respectively, then the region of space may have :

(a) $E=0, B=0$ (b) $E=0, B\neq0$ (c) $E\neq0, B=0$ (d) $E\neq0, B\neq0$

Choose the most appropriate answer from the options given below :

A) (b), (c) and (d) only B) (a), (c) and (d) only

C) (a), (b) and (d) only D) (a), (b) and (c) only

15. A wire of length 10 cm and radius $\sqrt{7}\times10^{-4}$ m is connected across the right gap of a meter bridge. When a resistance of $4.5\ \Omega$ is connected on the left gap by using a resistance box, the balance length is found to be at 60 cm from the left end. If the resistivity of the wire is $R\times10^{-7}\Omega m$, then value of R is

A) 70 B) 63 C) 66 D) 35

16. The average kinetic energy of a monoatomic molecule is 0.414eV at temperature :

(Use $K_B = 1.38\times10^{-23} J/mol-K$)

A) 1500K B) 1600K C) 3200K D) 3000K

17. Two bodies of mass 4g and 25g are moving with equal kinetic energies. The ratio of magnitude of their linear momentum is

A) 5:4 B) 3:5 C) 2:5 D) 4:5

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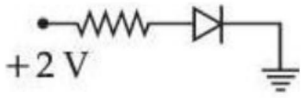
18. Position of an ant (S in metres) moving in Y-Z plane is given by $S=2t^2\hat{j}+5\hat{k}$ (where t is in second). The magnitude and direction of velocity of the ant at $t=1s$ will be

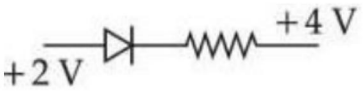
A) 4 m/s in y-direction
B) 9 m/s in z-direction
C) 16 m/s in y-direction
D) 4 m/s in x-direction

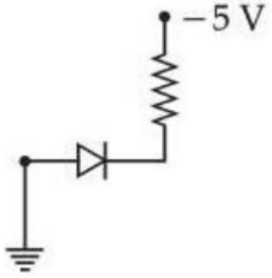
19. The radius of third stationary orbit of electron for Bohr's atom is R. The radius of fourth stationary orbit will be

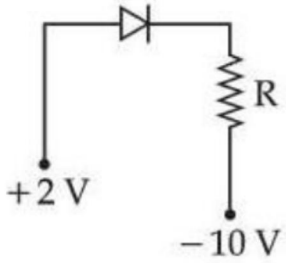
A) $\frac{4}{3}R$ B) $\frac{3}{4}R$ C) $\frac{9}{16}R$ D) $\frac{16}{9}R$

20. Which of the following circuits is reverse – biased?

A) 

B) 

C) 

D) 

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and −1 if not correct.

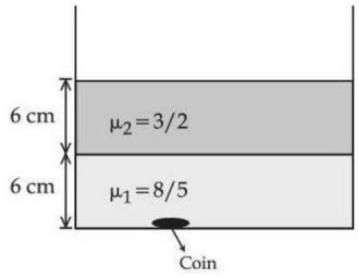
21. A thin metallic wire having cross sectional area of $10^{-4}m^2$ is used to make a ring of radius 30 cm. A positive charge of $2\pi C$ is uniformly distributed over the ring, while another positive charge of 30 pC is kept at the centre of the ring. The tension in the ring is _____ N; provided that the ring does not get deformed (neglect the influence of gravity). (given, $\frac{1}{4\pi\epsilon_0}=9\times10^9$ SI units)

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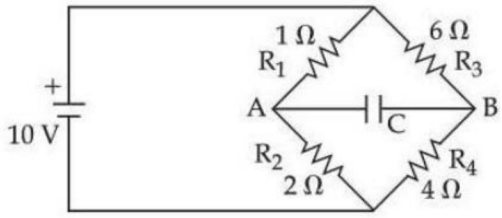
22. Four particles each of mass 1kg are placed at four corners of a square of side 2m. Moment of inertia of system about an axis perpendicular to its plane and passing through one of its vertex is _____ kgm^2.

23. Two immiscible liquids of refractive indices $\frac{8}{5}$ and $\frac{3}{2}$ respectively are put in a beaker as shown in the figure. The height of each column is $6\,\text{cm}$. A coin is placed at the bottom of the beaker. For near normal vision, the apparent depth of the coin is $\frac{\alpha}{4}\text{cm}$. The value of α is _____.



24. The charge accumulated on the capacitor connected in the following circuit is _____ μC.

(Given : $C=150\mu\text{F}$)



25. A particle starts from origin at $t=0$ with a velocity $5\hat{i}\ \text{m/s}$ and moves in x-y plane under action of a force which produces a constant acceleration of $(3\hat{i}+2\hat{j})\ \text{m/s}^2$. If the x-coordinate of the particle at that instant is $84\ \text{m}$, then the speed of the particle at this time is $\sqrt{\alpha}\ \text{m/s}$. The value of α is _____.

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CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. Given below are two statements :

Statement (I) : p-nitrophenol is more acidic than m-nitrophenol and o-nitrophenol.

Statement (II) : Ethanol will give immediate turbidity with Lucas reagent.

In the light of the above statements, choose the correct answer from the options given below :

A) Statement I is true but Statement II is false

B) Both Statement I and Statement II are false

C) Both Statement I and Statement II are true

D) Statement I is false but Statement II is true

27. Choose the polar molecule from the following :

A) CCl_4 B) CO_2 C) $CH_2=CH_2$ D) $CHCl_3$

28. Which of the following electronic configuration would be associated with the highest magnetic moment?

A) $[Ar]3d^3$ B) $[Ar]3d^6$ C) $[Ar]3d^7$ D) $[Ar]3d^8$

29. Consider the following complex ions

$P=[FeF_6]^{3-}$, $Q=[V(H_2O)_6]^{2+}$, $R=[Fe(H_2O)_6]^{2+}$

The correct order of the complex ions, according to their spin only magnetic moment values (in B.M.) is :

A) $Q<R<P$ B) $R<Q<P$ C) $R<P<Q$ D) $Q<P<R$

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30. The correct statement regarding nucleophilic substitution reaction in a chiral alkyl halide is

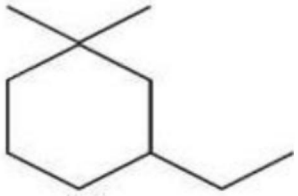
A) Racemisation occurs in S_N1 reaction and inversion occurs in S_N2 reaction

B) Retention occurs in S_N1 reaction and inversion occurs in S_N2 reaction

C) Racemisation occurs in S_N1 reaction and retention occurs in S_N2 reaction

D) Racemisation occurs in both S_N1 and S_N2 reactions

31. IUPAC name of following compound (P) is :



(P)

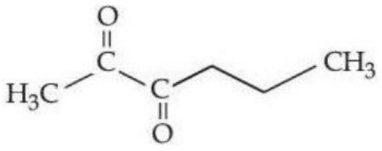
A) 1,1-Dimenthyl-3-ethylcyclohexane

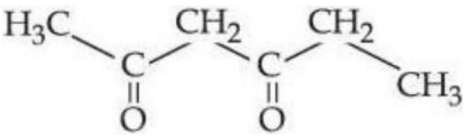
B) 1-Ethyl-5,5-dimethylcyclohexane

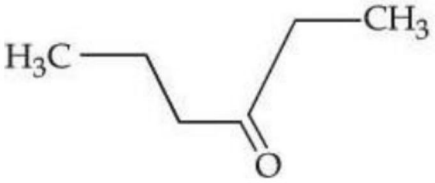
C) 3-Ethyl-1,1-dimethylcyclohexanane

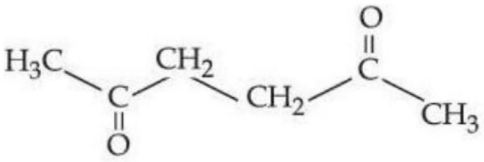
D) 1-Ethyl-3,3-dimethylcyclohexane

32. Which of the following has highly acidic hydrogen?

A) 

B) 

C) 

D) 

33. Given below are two statements :

Statement (I) : Aqueous solution of ammonium carbonate is basic.

Statement (II) : Acidic/basic nature of salt solution of a salt of weak acid and weak base depends on K_a and K_b value of acid and the base forming it.

In the light of the above statements, choose the most appropriate answer from the options given below :

A) Statement I is incorrect but Statement II is correct

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B) Both Statement I and Statemeent II are correct

C) Both Statement I and Statement II are incorrect

D) Statement I is correct but Statement II is incorrect

34. Cyclohexene



is _____ type of an organic compound.

A) Benzenoid non-aromatic B) Benzoid aromatic

C) Acyclic D) Alicyclic

35. NaCl reacts with conc. H_2SO_4 and $K_2Cr_2O_7$ to give reddish fumes (B), which react with NaOH to given yellow solution (C). (B) and (C) respectively are :

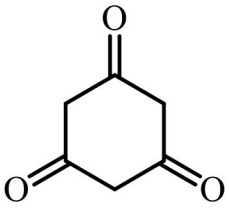
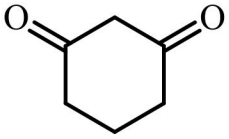
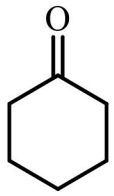
A) Na_2CrO_4, CrO_2Cl_2 B) $CrO_2Cl_2, Na_2Cr_2O_7$ C) $CrO_2Cl_2, KHSO_4$ D) CrO_2Cl_2, Na_2CrO_4

36. Two nucleotides are joined together by a linkage known as :

A) Disulphide linkage B) Peptide linkage

C) Glycosidic linkage D) Phosphodiester linkage

37. Highest enol content will be shown by :

A)  B)  C)  D) 

38. Yellow compound of lead chromate gets dissolved on treatment with hot NaOH solution. The product of lead formed is a :

A) Dianionic complex with coordination number four

B) Neutral complex with coordination number four

C) Tetraanionic complex with coordination number six

D) Dianionic complex with coordination number six

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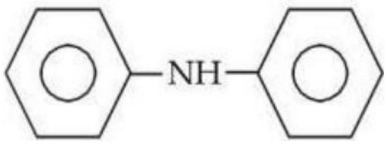
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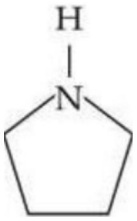
39. The electronic configuration for Neodymium is:

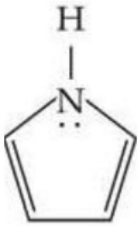
[Atomic Number for Neodymium 60]

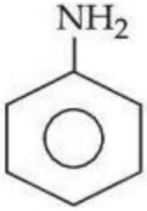
A) $[Xe]5f^7 7s^2$ B) $[Xe]4f^4 6s^2$ C) $[Xe]4f^1 5d^1 6s^2$ D) $[Xe]4f^6 6s^2$

40. Which of the following is strongest Bronsted base?

A) 

B) 

C) 

D) 

41. Element not showing variable oxidation state is :

A) Chlorine B) Iodine C) Bromine D) Fluorine

42. A solution of two miscible liquids showing negative deviation from Raoult's law will have :

A) Increased vapour pressure, increased boiling point

B) Decreased vapour pressure, increased boiling point

C) Decreased vapour pressure, decreased boiling point

D) Increased vapour pressure, decreased boiling point

43. Given below are two statements :

Statement (I) : The 4f and 5f - series of elements are placed separately in the periodic table to preserve the principle of classification.

Statement (II) : s-block elements can be found in pure form in nature.

In the light of the above statements, choose the most appropriate answer from the options given below :

A) Statement I is true but Statement II is false

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B) Statement I is false but Statement II is true

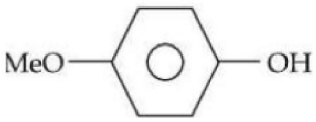
C) Both Statement I and Statement II are false

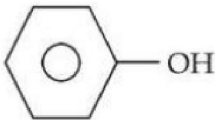
D) Both Statement I and Statement II are true

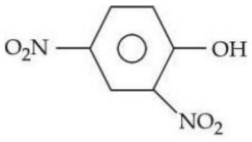
44. The ascending order of acidity of –OH group in the following compounds is :

(a) Bu – OH

(b) 

(c) 

(d) 

(e) 

A) (a) < (c) < (d) < (b) < (e)

B) (c) < (d) < (b) < (a) < (e)

C) (a) < (d) < (c) < (b) < (e)

D) (c) < (a) < (d) < (b) < (e)

45. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Melting point of Boron (2453 K) is unusually high in group 13 elements.

Reason (R) : Solid Boron has very strong crystalline lattice.

In the light of the above statements, choose the most appropriate answer from the options given below :

A) Both (A) and (R) are correct and (R) is the correct explanation of (A)

B) (A) is false but (R) is true

C) (A) is true but (R) is false

D) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. The mass of silver (Molar mass of Ag : 108 $gmol^{-1}$) displaced by a quantity of electricity which displaces 5600mL of O_2 at S.T.P. will be _____ g.

47. Consider the following data for the given reaction

$2\,HI_{(g)} \rightarrow H_{2(g)} + I_{2(g)}$

	1	2	3
HI (mol L^{-1})	0.005	0.01	0.02
Rate (mol $L^{-1}s^{-1}$)	7.5×10^{-4}	3.0×10^{-3}	1.2×10^{-2}

The order of the reaction is _____.

48. 3-Methylhex-2-ene on reaction with HBr in presence of peroxide forms an addition product (A).

The number of possible stereoisomers for 'A' is _____.

49. The number of electrons present in all the completely filled subshells having $n=4$ and $s=+\frac{1}{2}$ is _____.

(Where n = principal quantum number and s = spin quantum number)

50. If three moles of an ideal gas at 300K expand isothermally from $30dm^3$ to $45dm^3$ against a constant opposing pressure of 80 kPa , then the amount of heat transferred is _____ J.

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MATHS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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51. ${}^{n-1}C_r = (k^2-8)^n C_{r+1}$ if and only if :

A) $2\sqrt{3} < k \le 3\sqrt{2}$ B) $2\sqrt{3} < k < 3\sqrt{3}$

C) $2\sqrt{2} < k < 2\sqrt{3}$ D) $2\sqrt{2} < k \le 3$

52. If A denotes the sum of all the coefficients in the expansion of $(1-3x+10x^2)^n$ and B denotes the sum of all the coefficients in the expansion of $(1 + x^2)^n$, then

A) $B = A^3$ B) $3A = B$ C) $A = 3B$ D) $A = B^3$

53. If the shortest distance between the lines $\frac{x-4}{1} = \frac{y+1}{2} = \frac{z}{-3}$ and $\frac{x-\lambda}{2} = \frac{y+1}{4} = \frac{z-2}{-5}$ is $\frac{6}{\sqrt{5}}$, then the sum of all possible value of λ is

A) 10 B) 8 C) 7 D) 5

54. The distance, of the point $(7,-2,11)$ from the line $\frac{x-6}{1} = \frac{y-4}{0} = \frac{z-8}{3}$ along the line $\frac{x-5}{2} = \frac{y-1}{-3} = \frac{z-5}{6}$, is

A) 14 B) 21 C) 12 D) 18

55. The portion of the line $4x+5y=20$ in the first quadrant is trisected by the lines L_1 and L_2 passing through the origin. The tangent of an angle between the lines L_1 and L_2 is

A) $\frac{2}{5}$ B) $\frac{25}{41}$ C) $\frac{8}{5}$ D) $\frac{30}{41}$

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56. Let $a_1, a_2, ...a_{10}$ be 10 observations such that $\sum_{k=1}^{10} a_k = 50$ and $\sum_{\forall k<j} a_k . a_j = 1100$. Then the standard deviation of $a_1, a_2, ..., a_{10}$ is equal to

A) $\sqrt{115}$ B) 10 C) $\sqrt{5}$ D) 5

57. The function $f : N - \{1\} \to N$; defined by $f(n) =$ the highest prime factor of n, is

A) One-one only B) Onto only

C) Both one-one and onto D) Neither one-one nor onto

58. Let $S = \{1, 2, 3, ..., 10\}$. Suppose M is the set of all the subsets of S, then the relation $R = \{(A, B) : A \cap B \neq \phi; A, B \in M\}$ is

A) Symmetric only B) Reflexive only

C) Symmetric and transitive only D) Symmetric and reflexive only

59. Let $x = x(t)$ and $y = y(t)$ be solutions of the differential equations $\frac{dx}{dt} + ax = 0$ and $\frac{dy}{dt} + by = 0$ respectively, a, $b \in R$. Given that $x(0) = 2$; $y(0) = 1$ and $3y(1) = 2x(1)$, the value of t, for which $x(t) = y(t)$, is

A) $\log_{\frac{2}{3}} 2$ B) $\log_3 4$ C) $\log_{\frac{4}{3}} 2$ D) $\log_4 3$

60. Consider the matrix $f(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Given below are two statements :

Statement I : $f(-x)$ is the inverse of the matrix $f(x)$.

Statement II : $f(x) f(y) = f(x + y)$.

In the light of the above statements, choose the correct answer from the options given below

A) Both Statement I and Statement II are false

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B) Both Statement I and Statement II are true

C) Statement I is false but Statement II is true

D) Statement I is true but Statement II is false

61. If the shortest distance of the parabola $y^2 = 4x$ from the centre of the circle $x^2 + y^2 - 4x - 16y + 64 = 0$ is d, then d^2 is equal to

A) 20 B) 24 C) 16 D) 36

62. The number of common terms in the progressions 4, 9, 14, 19, ….., up to 25th term and 3, 6, 9, 12, …., up to 37th term is :

A) 5 B) 7 C) 9 D) 8

63. Consider the function, $f(x) = \begin{cases} \dfrac{a(7x-12-x^2)}{b\left|x^2-7x+12\right|} & , \quad x<3 \\ 2^{\frac{\sin(x-3)}{x-[x]}} & , \quad x>3, \\ b & , \quad x=3 \end{cases}$

where $[x]$ denotes the greatest integer less than or equal to x. If S denotes the set of all ordered pairs (a,b) such that $f(x)$ is continuous at $x=3$, then the number of elements in S is

A) 1 B) 2 C) Infinitely many D) 4

64. If $\int_0^1 \frac{1}{\sqrt{3+x}+\sqrt{1+x}}dx = a + b\sqrt{2} + c\sqrt{3}$, where a, b, c are rational numbers, then $2a+3b-4c$ is equal to

A) 4 B) 10 C) 7 D) 8

65. If $S = \{z \in C : |z-i| = |z+i| = |z-1|\}$, then $n(S)$ is

A) 0 B) 2 C) 3 D) 1

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66. The length of the chord of the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$, whose mid point is $\left(1,\frac{2}{5}\right)$, is equal to

A) $\frac{\sqrt{1541}}{5}$ B) $\frac{\sqrt{1741}}{5}$ C) $\frac{\sqrt{2009}}{5}$ D) $\frac{\sqrt{1691}}{5}$

67. If (a,b) be the orthocenter of the triangle whose vertices are $(1,2),(2,3)$ and $(3,1)$, and $I_1=\int_a^b x\sin(4x-x^2)dx$, $I_2=\int_a^b \sin(4x-x^2)dx$, then $36\frac{I_1}{I_2}$ is equal to

A) 88 B) 72 C) 80 D) 66

68. If $a=\lim_{x\to 0}\frac{\sqrt{1+\sqrt{1+x^4}}-\sqrt{2}}{x^4}$ and $b=\lim_{x\to 0}\frac{\sin^2 x}{\sqrt{2}-\sqrt{1+\cos x}}$, then the value of ab^3 is

A) 25 B) 32 C) 30 D) 36

69. Let $\vec{a}=\hat{i}+2\hat{j}+\hat{k}$, $\vec{b}=3(\hat{i}-\hat{j}+\hat{k})$. Let $\vec{c}$ be the vector such that $\vec{a}\times\vec{c}=\vec{b}$ and $\vec{a}.\vec{c}=3$. Then $\vec{a}.((\vec{c}\times\vec{b})-\vec{b}-\vec{c})$ is equal to

A) 36 B) 24 C) 32 D) 20

70. Four distinct points $(2k,3k)$, $(1,0)$, $(0,1)$ and $(0,0)$ lie on a circle for k equal to_____.

A) $\frac{3}{13}$ B) $\frac{2}{13}$ C) $\frac{1}{13}$ D) $\frac{5}{13}$

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. A fair die is tossed repeatedly until a six is obtained. Let X denote the number of tosses required and let $a = P(X = 3)$, $b = P(X \geq 3)$ and $c = P(X \geq 6 / X > 3)$. Then $\frac{b+c}{a}$ is equal to _____.

72. Let the set of all $a \in R$ such that the equation $\cos 2x + a \sin x = 2a - 7$ has a solution be $[p, q]$ and $r = \tan 9° - \tan 27° - \frac{1}{\cot 63°} + \tan 81°$, then pqr is equal to _____.

73. Let for a differentiable function $f : (0, \infty) \to R, f(x) - f(y) \geq \log_e \left(\frac{x}{y}\right) + x - y, \ \forall x,$ $y \in (0, \infty)$. Then $\sum_{n=1}^{20} f'\left(\frac{1}{n^2}\right)$ is equal to _____.

74. Let the area of the region $\{(x, y) : x - 2y + 4 \geq 0, x + 2y^2 \geq 0, x + 4y^2 \leq 8, y \geq 0\}$ be $\frac{m}{n}$, where m and n are coprime numbers. Then $m + n$ is equal to _____.

75. If α satisfies the equation $x^2 + x + 1 = 0$ and $(1 + \alpha)^7 = A + B\alpha + C\alpha^2$, A, B, $C \geq 0$, then $5(3A - 2B - C)$ is equal to _____.

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